

Qatar-1b

R-0404

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180619 · created 2026-07-19T04:38:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458288.8784 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.056
Transit depth (Rp/Rs)^2	3.69 +/- 2.13 [%]
Orbital Inclination (inc)	89.05 +/- 2.84
Airmass coefficient 1 (a1)	0.89 +/- 0.36
Airmass coefficient 2 (a2)	0.05 +/- 1.36
Scatter in the residuals of the lightcurve fit is	38.58 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.0834 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.056 vs 0.1463 (deviation 0.79σ)
Duration fit vs published	0.0834 d vs 0.0692 d (deviation 2.06σ)
Mid-time O–C	-0.1 min
Depth SNR	3.3
Residual scatter	38.58 %

Light curve

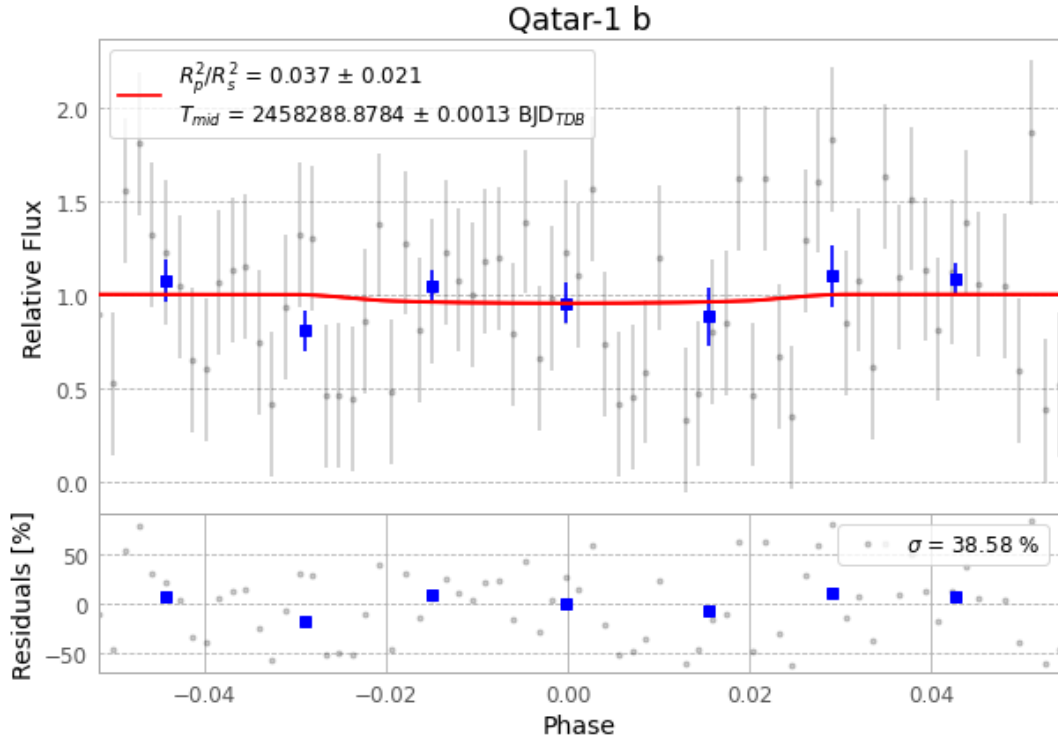


Figure 1 — FinalLightCurve_Qatar-1 b_2018-06-19.png (SHA-256 68d12a0e8bf7abf9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [132, 362], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8ca66440f143cc9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

73 FITS frames (48 MB), Qatar-1180619071812.FITS → Qatar-1180619105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180619.log (23b8f73a7d85...), FinalLightCurve_Qatar-1 b_2018-06-19.png (68d12a0e8bf7...), FinalLightCurve_Qatar-1 b_2018-06-19.csv (318f7246f6d1...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 04:38Z.